

ITAI 1371 Introduction to Machine Learning

Professor Viswanatha Rao (Vishwa)

Assignment: Lab 06

July 18, 2026

Reflection Journal

Completing this lab helped me better understand the difference between regression and classification models. Before this activity, I understood the basic definitions, but working through the notebook made the concepts clearer. I learned that regression is used to predict a continuous numerical value, while classification is used to predict a category or class. In this lab, predicting a passenger's fare was a regression problem, while predicting whether a passenger survived was a classification problem.

My individual contribution included completing and reviewing both machine-learning sections of the notebook. I trained the Linear Regression model using passenger age and class to predict fare, generated predictions, and evaluated the results using mean squared error. I also prepared the data for the Logistic Regression model, trained it to predict passenger survival, and evaluated its performance using accuracy. In addition, I added comments to explain the important coding steps and completed the knowledge-check questions in my own words.

One challenge I faced was understanding how to interpret the model results instead of only focusing on whether the code ran successfully. The regression model produced an RMSE of about \$58.01, indicating that predicted fares could differ noticeably from actual fares. The classification model achieved about 74.83% accuracy, meaning it correctly predicted survival for most passengers in the test set. My main takeaway from this lab is that choosing the correct model and evaluation metric is

important because regression and classification problems measure success in different ways.

In this journal entry, I will share my thoughts and experiences on Module 6 and how to train a linear regression model and a logistic regression model. This lab was designed to teach me about each of the models and how to properly identify whether to use one model over the other. This module taught me about linear regression vs logistic regression, where linear takes input to give numerical outcomes, while logistic regression takes input to classify the input into a certain category on a familiar dataset, which was used in Modules 4 and 5 to practice one hot encoding, and now taking that data and making models with it.

This module taught me about 2 different types of ways to train a machine learning model. The first way is linear regression. This method is great for predicting how a number will change based on different inputs. When I first read it in the PowerPoint presentation, I thought about playing a video game where you can change your character's weapon or what buffs you receive. Based on multiple factors like weapon level and what items you may equip, your damage will increase; damage taken from a boss may reduce your damage. This is how linear regression works: your damage will be affected by each buff and debuff it receives. This model is trained by mean squared error. This is a great loss function to train linear regression, as if you have the same negative and positive numbers, they would just cancel out, but with this method they don't as they square the numbers, so any negative becomes positive.

The other method I learned in this lab is logistic regression. This is very similar to linear regression, where you're taking multiple inputs to get an output; however, linear will give a numerical output while logistic will put the output into different categories. Think about the video game analogy from before: take it as a team based game now instead of a single-player game. If you switch your character, your entire role has a chance to change, such as if you switch to a healer, your entire purpose in the game changes. Instead of being in the front line and dealing damage, you are now behind the main damage dealer, helping them survive. This is how logistic regression works: it takes character status and, based on this information, can predict what category they go in, if a character has a high attack stat and low to medium health points or has dealt most of the team damage, a logistic regression model can see this and predict with 90% probability that this character is a Main damage dealer.

In this lab, I used both models for the Titanic dataset. I used linear regression to predict fare and the inputs I took were age and Pclass. It returned 3364 for the squared error which

means when we take the square root of this number, my model is off by about 58 dollars. A question I had at this point was that, in the print statement, after it says that the model should only be off by 48 dollars, I had tried different inputs to get to this number, but it kept returning an error. I would like to know if this is my fault or if this is just a typo, and if it is my fault, I would like to learn what to do next time about a problem like this. For logistic regression, I used age, Pclass, and sex to determine if a passenger survived or not. I got a 74% accuracy rating. After I learned how both models work, I realized that the same dataset can be used for very different outputs depending on how your model works and what it's trying to achieve.\

These lab methods, like before, are based on careful consideration of what to use to train these models and what not to train these models with. If not picked carefully, this could waste resources and time. This could happen if you use the wrong inputs for your model and it won't have a high accuracy rate. I hope to learn about the other methods of machine learning later on in this course, and how they differ from these 2 models.

Reflective Journal: Module 06 Lab - Regression and Classification

For Module 06, we focused on regression and classification. This lab helped me understand the difference between the two a lot better because we actually used both of them in code instead of just reading about them. Before doing the lab, I understood the basic definition, but working through the Titanic dataset made it more clear. Regression is used when we are trying to predict a number, and classification is used when we are trying to predict a category.

In the Linear Regression part, we used the model to predict the passenger fare. This made sense as a regression problem because fare is a number and can have many different values. The model was not trying to say yes or no. It was trying to estimate an amount. Looking at the mean squared error helped me understand that regression is not judged the same way as classification. With fare prediction, the model can be a little off, very off, or close enough, so we need a metric that measures the size of the error.

The Logistic Regression part was different because we used it to predict whether a passenger survived or not. Even though the name has “regression” in it, this part helped me remember that Logistic Regression is really used for classification. The output is more like a category, such as survived or did not survive. Accuracy made more sense here because the prediction is either correct or incorrect.

One thing I liked about this lab was looking at the coefficients. They helped me see that the model is not just giving random predictions. The coefficients give an idea of how each feature is affecting the result. I still do not feel like I understand every detail perfectly, but I understand the general idea better now. A positive coefficient means the feature pushes the prediction upward, while a negative coefficient pushes it downward.

The hardest part for me was keeping the two sections separate in my head. Since both sections used the Titanic dataset, it was easy to mix up what we were predicting. I had to remind myself that predicting `Fare` was regression, while predicting `Survived` was classification. Once I slowed down and focused on the target variable, the difference made more sense.

This lab also reminded me that data preparation still matters. Even though this module was about models, the data still had to be cleaned and converted into a format the model could use. For example, the `Sex` column had to be turned into numbers before Logistic Regression could use it. That connected back to what we learned in the earlier data preparation module.

Overall, this lab helped me understand supervised learning better. I learned that the type of target variable decides which kind of model and evaluation metric should be used. If the target is a number, regression makes sense. If the target is a category, classification makes sense. This was one of the clearer labs for me because I could see the difference directly in the code and outputs.